# Martina Contisciani

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## Current Position

**Postdoctoral Researcher** May 2026 – Present

*Center for Critical Computational Studies - Inverse Complexity Lab, Frankfurt am Main, Germany*

- Research focused on the development of statistical methods and algorithms to analyze complex systems, with a particular interest in network inference.

**Freelancer Collaborator** May 2026 – Present

*Central European Univerisity - Dept. of Network and Data Science, Vienna, Austria*

- Work related to the MOMA (Multiscale network modelling of migration flows in Austria) project.

## Experience

**Postdoctoral Researcher** Mar. 2024 – Apr. 2026

*Central European Univerisity - Dept. of Network and Data Science, Vienna, Austria*

- Research focused on the analysis of the multiscale network structure of migration flows in Austria as part of the MOMA project, together with Prof. Tiago P. Peixoto and Prof. Márton Karsai

**Teaching Assistant** Nov. 2019 – Feb. 2022

*University of Tübingen, Tübingen, Germany*

- TA for Advanced Probabilistic Machine Learning and Applications (M.Sc) - WS 2019/2020, 2021/2022, SS 2021
- TA for Statistical Machine Learning (M.Sc) - SS 2020

**Intern** Mar. 2019 – Sep. 2019

*Max Planck Institute for Intelligent Systems - PIO Group, Tübingen, Germany*

- Master's thesis project focused on modelling covariate information in community detection algorithms

**Data Analyst** Jul. 2018 – Aug. 2018

*Unox S.p.A. – IT Office, Cadoneghe (PD), Italy*

- Project focused on the analysis of device usage data to improve the experience and profile of the customers

## Education

**Ph.D. in Computer Science** Oct. 2019 – Nov. 2023

*Max Planck Institute for Intelligent Systems - PIO Group, Tübingen, Germany*

- Thesis: Probabilistic Generative Models for Inference on Complex Systems
- Final Grade: Summa cum laude

**Master's degree in Data Science** Oct. 2017 – Sep. 2019

*University of Padua - Dept. of Mathematics, Padua, Italy*

- Thesis: A New Approach for Community Detection in Multilayer Networks
- Final Grade: 110/110 cum laude

**Bachelor's degree in Statistics for Business and Economics** Oct. 2014 – Sep. 2017

*University of Padua - Dept. of Statistical Sciences, Padua, Italy*

- Thesis: Automatic identification of time series and forecasting performances
- Final Grade: 110/110 cum laude

## Publications

13. Robiglio T.*, **Contisciani M.***, Karsai M., and Peixoto P. T., (2026), Multiscale patterns of migration flows in Austria: regionalization, administrative barriers, and urban-rural divides, *Scientific Reports 2026* - *contributed equally

12. Baptista D., **Contisciani M.**, De Bacco C., and Passy J.-C., (2026), ProbINet: Bridging Usability Gaps in Probabilistic Network Analysis, *Journal of Open Source Software, 11(124), 8638*

11. **Contisciani M.***, Hobbhahn M.*, Power E. A., Hennig P., and De Bacco C., (2025), Flexible inference in heterogeneous and attributed multilayer networks, *PNAS Nexus, pgaf005* - *contributed equally

10. Safdari H., **Contisciani M.**, De Bacco C., (2023), Anomaly, reciprocity, and community detection in networks, *Phys. Rev. Res. 5, 033084*

9. Ruggeri N., **Contisciani M.**, Battiston F., De Bacco C., (2023), Community detection in large hypergraphs, *Science Advances 9, eadg9159*

8. Lotito Q. F., **Contisciani M.**, De Bacco C., Di Gaetano L., Gallo L., Montresor A., Musciotto F., Ruggeri N., Battiston F., (2023), Hypergraphx: a library for higher-order network analysis, *Journal of Complex Networks 11, cnad019*

7. De Bacco C., **Contisciani M.**, Cardoso-Silva J., Safdari H., Baptista D., Borges G. L., Sweet T., Young J.-G., Koster J., Ross C. T., McElreath R., Redhead D., Power E. A., (2023), Latent network models to account for noisy, multiply reported social network data, *Journal of the Royal Statistical Society Series A: Statistics in Society, qnac004*

6. **Contisciani M.**, Battiston F., De Bacco C., (2022), Inference of hyperedges and overlapping communities in hypergraphs, *Nature Communications 13, 7229*

5. **Contisciani M.**, Safdari H., De Bacco C., (2022), Community detection and reciprocity in networks by jointly modelling pairs of edges, *Journal of Complex Networks 10, cnac034*

4. Higham K., **Contisciani M.**, De Bacco C., (2022), Multilayer patent citation networks: A comprehensive analytical framework for studying explicit technological relationships, *Technological Forecasting and Social Change 179, 121628*

3. Safdari H., **Contisciani M.**, De Bacco C., (2022), Reciprocity, community detection, and link prediction in dynamic networks, *Journal of Physics: Complexity 3, 015010*

2. Safdari H.*, **Contisciani M.***, De Bacco C., (2021), Generative model for reciprocity and community detection in networks, *Phys. Rev. Res. 3, 023209* - *contributed equally

1. **Contisciani M.**, Power E. A., De Bacco C., (2020), Community detection with node attributes in multilayer networks, *Scientific Reports 2020 10:1 10, 1–16*

## Presentations

### Invited presentations, guest lectures, and seminars

[2026] *A Bridge from Collective Motion to Social Behavior, Migration, and Collective Decisions*, Vienna (AT) - *IDWSDS 2026*, online - *COMPSTAT 2026*, Athens (GR) - *Dept. of Information Technology at UU*, Uppsala (SE) - *Seminar at QMUL*, London (UK) - *Vienna Workshop on Migration Modeling*, Vienna (AT)

[2025] *POPJUS Guest Seminar at IIASA*, Laxenburg (AT) - *Dept. of Network and Data Science at CEU*, Vienna (AT)

[2024] *2024 ISBA World Meeting*, Venice (IT)

[2023] *Young Seminars SIFS*, online

[2022] *NetPLACE*, online - *Dept. of Statistical Sciences at UNIPD*, Padua (IT)

[2021] *Dept. of Statistical Sciences at UNIPD*, online

## Contributed presentations

[2026] *BAYSM 2026*, Chiba (JP)

[2025] *NetSci 2025*, Maastricht (NL)

[2023] *NetSci 2023*, Vienna (AT) - *EUSN 2023*, Ljubljana (SI)

[2022] *NetSci 2022*, online - *EUSN 2022*, London (UK) - *CCS 2022*, Palma de Mallorca (ES)

[2021] *Networks 2021*, online - *EUSN 2021*, online

## Poster presentations

[2026] *2026 ISBA World Meeting*, Nagoya (JP)

[2020] *NetSci 2020*, online

## Service

**Open Source Contributor**: ProbINet, hypergraphx (HGX), VIMuRe, complex group interactions (XGI)
**Reviewer**: Applied Network Science, Communications Physics, Journal of Complex Networks, Journal of Computational and Applied Mathematics, Journal of the Royal Statistical Society Series A, New Journal of Physics, Social Network Analysis and Mining, Statistics and Computing
**Organizer**: ELLIS Doctoral Symposium 2021, StatisticALL 2016

## Awards

[2026] **Best Poster Prize**, *2026 ISBA World Meeting, Nagoya, Japan*

[2026] **BAYSM 2026 Travel Award**, *BAYSM 2026, Chiba, Japan*

[2023] **MPI-IS Outstanding Female Doctoral Student Prize**: honorable mention, *Max Planck Institute for Intelligent Systems, Tübingen, Germany*

[2018] **Hackathon about Speech Recognition**: 1st place position, *Unox S.p.A., Cadoneghe, Italy*

## Skills

**Languages**: Italian (mother tongue), English (upper-intermediate), German (intermediate)
**Soft Skills**: Conscientiousness, Agreeableness, Teamwork, Problem-Solving
**Technologies**: Python, R, MATLAB, SQL, TensorFlow, Keras, LaTeX, C, C#, HTML/CSS, Gephi, JDemetra+

**Last update**: October 12, 2026